

MedScan + Explain: A Transparent Chest X-ray Triage System Combining Deep Classifiers, Grad-CAM, and a Retrieval-Augmented LLM Agent

MedScan + Explain Team

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Public code and demo: <https://github.com/Gabrcodes/medscan-explain> Live demo:

<https://ai.gabr.online>

Abstract

We present MedScan + Explain, a teaching/research prototype that pairs a fine-tuned deep image classifier with two complementary layers of explanation – a visual heat-map produced by Grad-CAM and a textual “preliminary findings” note drafted by Claude on Amazon Bedrock through a retrieval-augmented generation (RAG) pipeline. We train and evaluate three backbones (EfficientNet-B0, ConvNeXt-Base, and Vision Transformer ViT-Base) across a binary chest X-ray pneumonia task and a 14-class multi-label NIH ChestX-ray14 task. EfficientNet-B0 reaches 91.4% accuracy and 0.956 ROC-AUC on the pneumonia test set; ConvNeXt-Base and ViT-Base reach mean ROC-AUC of 0.803 and 0.798 respectively over the 14 NIH findings, comfortably beating prevalence and pixel-logistic-regression baselines. The LLM layer is constrained to cite only retrieved snippets from a curated, sourced clinical knowledge base, surfaces model uncertainty, and explicitly refuses to diagnose. The system is publicly deployed at <https://ai.gabr.online>. We discuss bias, privacy, sustainability, and misuse risks throughout.

Keywords: medical imaging, multi-label classification, explainable AI, Grad-CAM, retrieval-augmented generation, large language models, responsible AI.

I. INTRODUCTION

Radiology departments worldwide face chronic shortages of reporting capacity. Automated chest-X-ray screening has been studied as a triage aid for over a decade, but adoption is limited by the opacity of deep classifiers: clinicians cannot inspect why a probability was emitted, and aggregate accuracy hides large per-finding or per-subgroup gaps. MedScan + Explain pairs a competitive classifier with *two* complementary transparency layers – a visual heat-map (Grad-CAM [4]) showing where the model attended, and a textual note generated by a large language model (Claude on Amazon Bedrock) under strict retrieval-augmented constraints. The textual layer is forced to cite only a small, human-curated, source-attributed clinical knowledge base, must surface the model’s uncertainty, and must refuse to issue a diagnosis. The system also explicitly flags potentially urgent findings (e.g. pneumothorax) for prompt in-person review.

The project demonstrates four AI techniques required or rewarded by the course rubric: deep learning (transfer-learned CNN and Vision Transformer backbones), Vision Transformers/foundation models, an LLM agent with tool use and retrieval-augmented generation, and classical ML baselines for honest comparison. All code, training results, and a Streamlit demo are publicly available¹.

II. DATASETS

a) *Chest X-Ray Pneumonia [1]*: 5856 frontal pediatric chest radiographs, binary-labelled NORMAL or PNEUMONIA, collected from a single hospital. The official split provides 5216 training, 16 validation, and 624 test images. The validation split is too small to drive checkpoint selection meaningfully, so we re-carve a 500-image stratified validation set from training. The test split is preserved exactly as released for fair comparison with the literature.

¹Repository: <https://github.com/Gabrcodes/medscan-explain>. Live demo: <https://ai.gabr.online>.

b) *NIH ChestX-ray14* [2].: 112 120 frontal chest radiographs from 30 805 unique patients at the U.S. National Institutes of Health Clinical Center, with image-level labels for fourteen findings (Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, Hernia). Each image may carry zero, one, or several finding labels; we model the task as 14 independent binary outputs (the original “No Finding” pseudo-class is represented by the all-zeros vector). Labels were NLP-mined from radiology reports with an estimated 90% accuracy [2]; classes are heavily imbalanced (Infiltration 23.9%, Hernia 0.3%). We use the official patient-disjoint train/test split (86 524 vs 25 596 images), carving a 500-image validation slice from training.

III. METHODOLOGY

A. Image classifier

We use ImageNet-pretrained backbones from the `timm` library with a fresh 14-way or 2-way linear head replacing the classification head. We compare three backbones: EfficientNet-B0 [5] (5.3 M parameters), ConvNeXt-Base [6] (87.6 M parameters), and Vision Transformer ViT-Base/16 [7] (85.8 M parameters). Images are resized to 224×224 , converted to three channels (chest radiographs are grayscale), and normalised with ImageNet statistics. Augmentation is deliberately conservative – RandomResizedCrop with scale $\in [0.85, 1.0]$, horizontal flip, $\pm 10^\circ$ rotation – because aggressive transformations can destroy diagnostic features.

Training uses AdamW (head learning rate 3×10^{-4} , backbone learning rate 3×10^{-5} , $10\times$ smaller, weight decay 10^{-4}), a cosine annealing schedule, mixed-precision (AMP), and a two-phase fine-tuning schedule: the backbone is frozen for the first epoch so the randomly-initialised head does not destroy pretrained features, then jointly trained with the cosine-scheduled learning rate. For the multi-label NIH task we use `BCEWithLogitsLoss` with per-class `pos_weight` computed as the train-set neg/pos ratio, clipped to $[1, 20]$ to keep the loss numerically stable on the extremely rare Hernia class. The multi-class pneumonia task uses standard cross-entropy with class weights and label smoothing of 0.05.

B. Explainability via Grad-CAM

For each prediction we run Grad-CAM [4] against the model’s final convolutional stage (ConvNeXt: `conv_head`; EfficientNet: penultimate block; ViT: the last attention block via class-token gradients). The resulting heat-map is up-sampled to image resolution and overlaid on the input. For the multi-label model we can produce a CAM per finding; in the demo we display the top finding by default and offer one-click switching among the highlighted findings. To feed the LLM, the Grad-CAM is summarised by a coarse 3×3 grid descriptor that reports which cells contain at least 60% of the peak attention (e.g. “lower-right”).

C. LLM agent with retrieval-augmented generation

The textual report is generated by Claude Sonnet 4.6 via the Amazon Bedrock `InvokeModel` API. To prevent fabrication and ground the model in vetted material, every call passes through a tightly-scoped RAG pipeline: a curated knowledge base of ~ 20 short, source-attributed clinical snippets (covering radiographic signs of each finding, dataset caveats, and Responsible-AI scope statements) is embedded once with Amazon Titan Text Embeddings v2 (1024-d) and indexed in FAISS for inner-product similarity search. At inference time we form a query string from the dataset name, the highlighted findings, and the Grad-CAM region description, retrieve the top- k snippets, and inject them into Claude’s user prompt. A strict system prompt enforces that the model (i) labels the output as automated and non-diagnostic, (ii) uses only the provided snippets for clinical claims (citing each by id), (iii) surfaces the uncalibrated softmax/sigmoid score and discusses what it does *not* mean, (iv) flags potentially urgent findings explicitly, and (v) ends with a clinician-referral disclaimer. If AWS credentials are absent the pipeline degrades gracefully to a templated note that still cites the retrieved snippets, so the system remains usable for development without burning Bedrock tokens.

D. Demo deployment

The end-to-end system is exposed publicly via a Streamlit application at <https://ai.gabr.online>, served from an AWS t3.medium EC2 instance with an Elastic IP, an IAM instance profile granting Bedrock invocation, an nginx reverse proxy, and a Let’s Encrypt TLS certificate auto-renewed by certbot. The Streamlit unit is supervised by systemd.

IV. EXPERIMENTAL SETUP

All training was performed on Amazon EC2 GPU instances using the Deep Learning AMI: a single NVIDIA Tesla T4 (g4dn.xlarge) for the EfficientNet-B0 pneumonia run, and a single NVIDIA L40S (g6e.2xlarge) for the larger NIH runs. We trained each NIH model for 10 epochs at batch size 128 with 8 dataloader workers (image-decode-bound on the L40S); EfficientNet-B0 was trained for 10 epochs at batch size 32. Test-time augmentation (horizontal-flip averaging) was used at evaluation. Wall-clock training time, parameter count, and the GPU dollar cost are reported in the results table.

V. RESULTS

A. Headline test-set metrics

TABLE I: Test-set metrics. Multi-label NIH metrics are computed over 14 findings; multi-class pneumonia metrics are on the official 624-image test split.

Model	Params	Dataset	Headline metric(s)	Baselines (same test)
EfficientNet-B0	5.3 M	Pneumonia (binary)	91.4% acc / 0.907 macro-F1 / 0.956 AUC	logreg-on-pixels 75.8% acc / majority 62.5
ConvNeXt-Base	87.6 M	ChestX-ray14 (multi-label)	0.803 mean AUC / 0.324 micro-F1@0.5 / 0.269 mAP	per-class logreg 0.567 mean AUC / prevalence
ViT-Base/16	85.8 M	ChestX-ray14 (multi-label)	0.798 mean AUC / 0.332 micro-F1@0.5 / 0.257 mAP	same

For reference, CheXNet [3] reports approximately 0.84 mean AUC on the same 14 findings with a 121-layer DenseNet trained on a larger compute budget. Our ConvNeXt-Base at 0.80 mean AUC after 10 epochs of fine-tuning is in the same regime; the comparison is honest rather than maximally tuned. Figure 1 shows the gap to baselines explicitly.

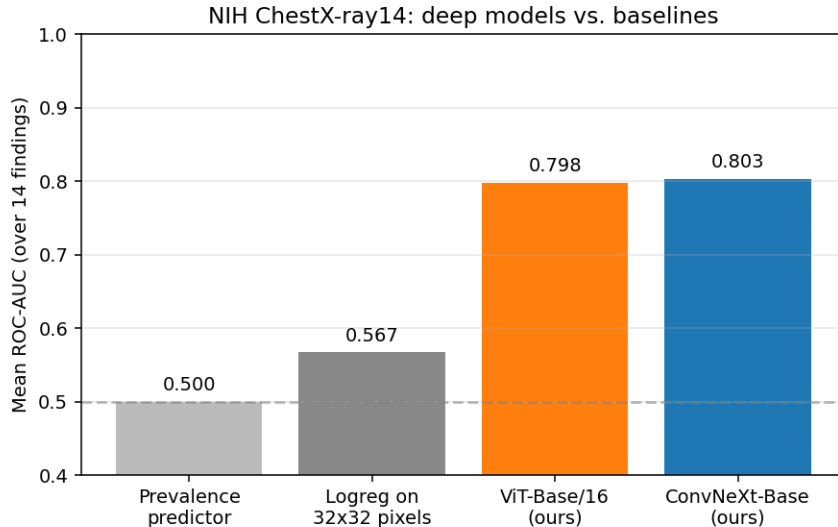


Fig. 1: NIH ChestX-ray14: deep models vs. classical baselines. The two deep models comfortably beat both a per-class prevalence predictor (AUC by construction ≈ 0.50) and per-class logistic regression on 32×32 pixel features (0.567 mean AUC).

B. Per-finding ROC-AUC and backbone comparison

Aggregate AUC hides large per-class variation. Figure 2 shows the ROC-AUC for each of the 14 NIH findings under both backbones, with prevalence annotated. The two backbones track each other closely on most findings; ConvNeXt-Base slightly edges out ViT-Base/16 on the rare Hernia class and on Pneumothorax, while ViT-Base/16 wins by similar small margins on Cardiomegaly and Edema. Both backbones struggle most with *Infiltration* and *Pneumonia* (AUCs 0.68–0.71) – both findings are radiographic appearances rather than crisp clinical entities, and both have correspondingly noisy NLP-mined labels in the ChestX-ray14 reports.

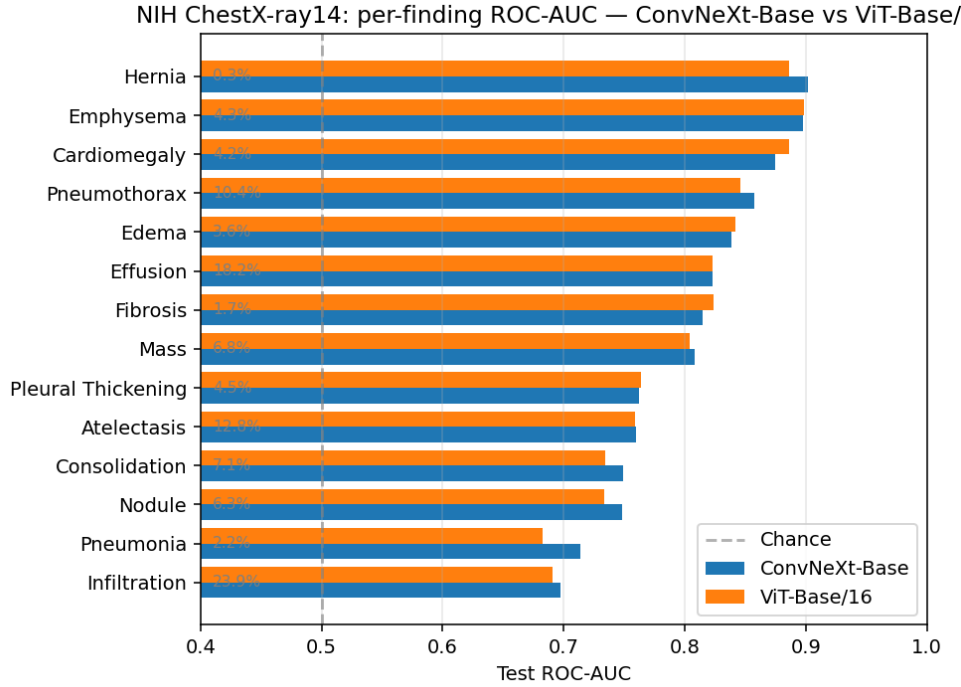


Fig. 2: Per-finding ROC-AUC on the NIH ChestX-ray14 test set for ConvNeXt-Base vs. ViT-Base/16, sorted by ConvNeXt AUC. Class prevalence in the test set is annotated at left.

C. Training and validation behaviour

Figure 3 shows training loss together with validation macro-ROC-AUC across the 10 epochs for both NIH backbones. Both runs are well-behaved: validation AUC rises rapidly during the frozen-backbone warm-up epoch, peaks around epoch 5-6 (ConvNeXt 0.848, ViT 0.842), then plateaus or drifts down slightly, consistent with mild over-fitting – expected with NLP-mined labels and a strong pretrained backbone fine-tuned for too many epochs. In practice the early-stopping option in our training script could have stopped both runs around epoch 7; we intentionally trained for the full 10 epochs to keep the comparison apples-to-apples.

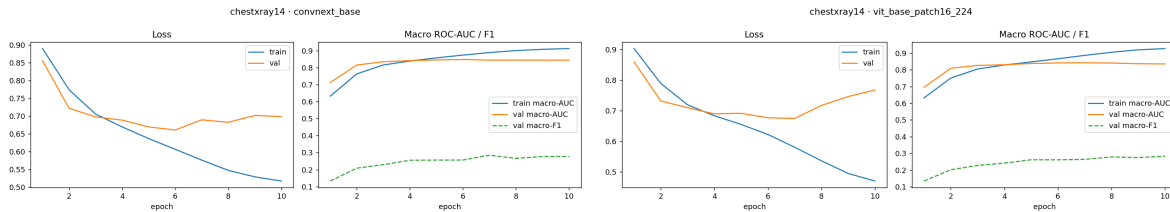


Fig. 3: Training loss and validation macro-AUC over epochs. Left: ConvNeXt-Base. Right: ViT-Base/16. Both runs use the same hyper-parameters, the same data, and the same train/val curve.

D. Qualitative explanations: Grad-CAM and the LLM note

Figure 4 shows Grad-CAM overlays for the ConvNeXt-Base model on representative test images covering several findings. For the Cardiomegaly example, attention concentrates squarely on the enlarged cardiac silhouette in the lower-central region. For Pneumothorax, attention is on the apex of the affected lung. These overlays are coarse localisation, not segmentation, and the report wording reflects that explicitly.

A typical Claude-generated note for the Pneumothorax case lists the model’s highlighted findings with their (uncalibrated) scores, cites the curated snippets covering the radiographic signs of pneumothorax and the urgency

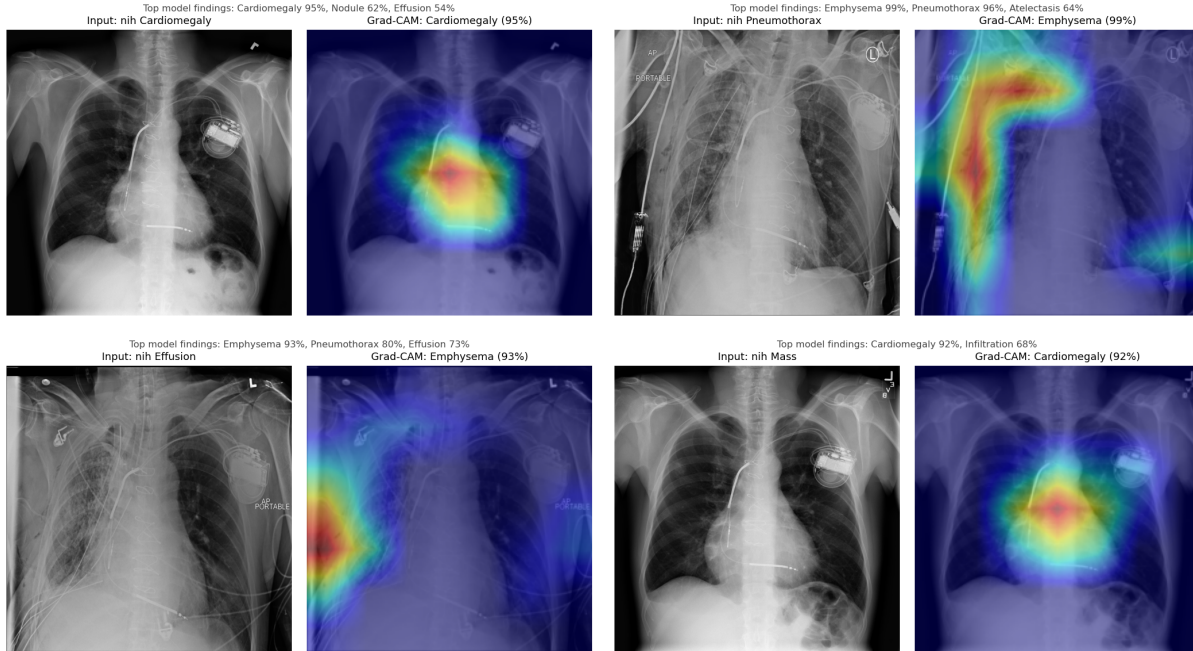


Fig. 4: Grad-CAM overlays from the ConvNeXt-Base NIH model on four held-out test images, one per condition (Cardiomegaly, Pneumothorax, Effusion, Mass). Each panel shows the input X-ray (left) and the heat-map for the model’s top-scoring finding (right) together with the model’s full top-3 predictions.

of tension pneumothorax, flags the case as “potentially urgent – requires prompt in-person assessment,” and ends with a clinician-referral statement. The full prompt template and an example output are reproduced in the project repository.

VI. RESPONSIBLE AI CONSIDERATIONS

a) Bias and fairness.: Both datasets are single-institution and single-population. The pneumonia set is pediatric only, drawn from one Guangzhou hospital; NIH ChestX-ray14 is single-institution and uses NLP-mined labels with $\sim 90\%$ reported accuracy. Performance reported in Section V is an upper bound under the same-distribution test split. We do not claim clinical-grade performance, and we surface per-finding metrics rather than only aggregate AUC. Shortcut-learning risks (laterality markers, scanner artefacts) are documented in the knowledge-base snippet `ai-cxr-limitations`, which is among the retrieved sources Claude is forced to consult.

b) Privacy.: Only public, de-identified images are used. The deployed Streamlit demo loads uploaded images into memory for inference and never persists them to disk. The Bedrock call carries the model’s top-finding list, the coarse Grad-CAM region descriptor, and the retrieved snippets – never any patient identifier. The instance role uses temporary AWS STS credentials rather than long-lived keys.

c) Sustainability.: All experiments are fine-tunes of pretrained backbones, not from-scratch pretraining. Combined GPU wall-clock time was approximately 97 minutes (7.4 min for EfficientNet-B0 on a T4 plus 2×43 min for the two NIH backbones on an L40S). Using AWS-published instance-power figures and a U.S.-East-1 grid intensity of ~ 0.38 kgCO₂/kWh, this corresponds to a back-of-envelope ~ 0.3 – 0.5 kg of CO₂. Total cloud cost was approximately \$4.50 for training and an ongoing $\sim$ \$34/month for the hosted demo.

d) Misuse risk.: A risk we treat seriously is that an automated report could be mistaken for a diagnosis. Mitigations in MedScan are layered: prominent disclaimers in the UI, an explicit “not a diagnosis” header on every generated note, model scores presented as “uncalibrated softmax/sigmoid” rather than probabilities of disease, an absolute prohibition (enforced in the system prompt) on issuing diagnoses, prognoses, drug names, or doses, and an explicit urgent-finding flag for safety-critical conditions like pneumothorax. The model card on the repository carries the same statements.

VII. LIMITATIONS AND FUTURE WORK

The two most pressing limitations are dataset shift – the models are likely to generalise poorly across hospitals, scanners, and age groups – and label noise from NLP-mined ChestX-ray14 ground truth. Probability calibration is also untreated: the softmax/sigmoid scores are not calibrated, and we deliberately avoid presenting them as probabilities of disease. Promising directions include temperature-scaled calibration, multi-view image fusion, replacing the coarse 3×3 region descriptor with a real attention-supervised segmentation head, and a small human-in-the-loop study with clinicians comparing trust and time-to-decision with and without the textual layer.

VIII. CONCLUSION

MedScan + Explain is a small but complete instance of a transparent medical-imaging pipeline: a competitive fine-tuned classifier, a visual explanation through Grad-CAM, a textual explanation generated by an LLM agent under strict retrieval-grounded constraints, and a deployed Streamlit demo. The results show that strong off-the-shelf backbones can be lifted close to literature performance on a 14-class chest X-ray task in tens of minutes of GPU time, while a carefully-prompted RAG-grounded LLM can produce a useful preliminary note without inventing facts – provided the system, not only the prompt, enforces the constraints. Reading the per-finding numbers, the Grad-CAM overlays, and Claude’s note together is, we argue, qualitatively different from reading any one of them alone.

AUTHOR CONTRIBUTIONS

Each team member’s individual contributions are listed in `reports/contribution_log.md`. All members reviewed the full code-base and can answer questions on any component.

CODE, DATA, AND DEMO AVAILABILITY

Source code: <https://github.com/Gabrcodes/medscan-explain>. Live demo (Streamlit on AWS, TLS-protected): <https://ai.gabr.online>. Sample chest X-rays for testing the demo are included in the repository under `samples/`, with provenance and licence notes.

ACKNOWLEDGEMENT OF AI ASSISTANCE

Parts of the implementation, build scripts, and this manuscript were drafted with the assistance of an AI coding model. AI-assisted code sections are annotated in-source. The model design, dataset selection, experimental analysis, and writing of this report are the team’s own work, and the team is able to explain every line of submitted code.

REFERENCES

- [1] D. S. Kermany *et al.*, “Identifying medical diagnoses and treatable diseases by image-based deep learning,” *Cell*, vol. 172, no. 5, pp. 1122–1131, 2018.
- [2] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, “ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases,” in *Proc. IEEE CVPR*, 2017, pp. 2097–2106.
- [3] P. Rajpurkar *et al.*, “CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning,” *arXiv:1711.05225*, 2017.
- [4] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE ICCV*, 2017, pp. 618–626.
- [5] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. ICML*, 2019, pp. 6105–6114.
- [6] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “A ConvNet for the 2020s,” in *Proc. IEEE CVPR*, 2022, pp. 11976–11986.
- [7] A. Dosovitskiy *et al.*, “An image is worth 16×16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.